

## CSV Gate Standard v1.2

Release: MathGov Core Release 2026.09 v11.4 Complete Ready

Source-boundary rule: If this compact standard conflicts with the RippleLogic Canon, the Canon controls.

CSV means Containment and Structural Viability. CSV is the strengthened fourth level of the RippleLogic cascade inside the MathGov framework:

RG -> RF -> TRC -> CSV -> RLS

RF means Rights Floor, the public-facing name for the Non-Compensatory Rights Constraint (NCRC). RF names the second cascade layer; NCRC names the formal pass/fail predicate for protected rights. NCRC-style notation remains valid in equations and checklists.

CSV is not a sixth public level. It is the upgraded fourth level: Contain/Verify.

TRC-CSV feedback rule. If CSV discovers a catastrophic, irreversible, or ruin-path scenario not represented in TRC, the run MUST reopen TRC before RLS. CSV does not absorb TRC.

Gate-critical confidence guard. Low confidence in adverse rights-covered, catastrophe-covered, or material CSV impacts cannot by itself make an option pass RF/NCRC, TRC, or CSV. If a gate outcome could change, the run must use a governed conservative bound, collect evidence and rerun, narrow the claim, downgrade, escalate, or refuse the stronger claim.

Baseline pointer. Gate-admissibility cells and residual welfare cells may use distinct baselines where the Canon requires it; see Canon §5.1A for the floor-reference versus status-quo dual-baseline rule.

## Realizability interface

CSV is the main structural-realizability layer, but it does not absorb Reality Grounding or TRC. RG decides whether the claim is grounded. TRC decides whether catastrophic tail exposure is bounded. CSV decides whether the pathway can stand under containment, dependency, execution, monitoring, authority, and host-system constraints. If CSV discovers a new catastrophic or ruin-path scenario, TRC must be reopened before RLS.

RG status rule. Reality Grounding is Level 1 of the public method and a claim-authority precondition. It can force narrowing, escalation, exploratory-marking, or refusal, but it is not an option-rejecting gate; RF/NCRC, TRC, and CSV are the option-rejecting gates, and RLS ranks only what survives them.

## CSV core rule

CSV does not demand zero harm. CSV demands that harms are visible, routed, bounded, mitigated, monitored, and not structurally degrading or unjustly externalized beyond tolerance.

## CSV status values

### CSV status resolver guide

The Canon controls, but this compact guide constrains routing:

| Evidence / diagnostic condition                                                                                   | CSV routing                                                               |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Any material diagnostic level 4 or uncontained / structurally degrading / unjustly externalized / non-viable harm | CSV_FAIL                                                                  |
| Material Structural Viability failure that is not controlled or redesignable                                      | CSV_FAIL                                                                  |
| Material Structural Viability failure that can plausibly be repaired                                              | CSV_REDESIGN_REQUIRED                                                     |
| Gate-material evidence missing, stale, contested, or insufficient                                                 | CSV_ESCALATE, claim narrowing, or refusal of the stronger selection claim |
| Material diagnostic level 3                                                                                       | CSV_REDESIGN_REQUIRED or CSV_ESCALATE                                     |
| Material diagnostic level 2 with binding controls                                                                 | CSV_PASS_WITH_CONTROLS                                                    |
| Material diagnostic level 2 without binding controls                                                              | CSV_REDESIGN_REQUIRED                                                     |
| All material diagnostics level 0-1 and Structural Viability passes                                                | CSV_PASS                                                                  |
| No material CSV burden under the declared claim boundary                                                          | CSV_NOT_MATERIAL                                                          |
| Necessity + no better feasible option + time limit + harm cap + monitoring + transition plan                      | CSV_EMERGENCY_PROVISIONAL                                                 |

Tier boundary. A CSV\_PASS or CSV\_PASS\_WITH\_CONTROLS in a Tier 2 worked run may be assumption-bound and must not be read as validated structural measurement.

UCI/HOI remain diagnostics until separate validation evidence supports stronger claims.

- CSV\_PASS
- CSV\_PASS\_WITH\_CONTROLS
- CSV\_REDESIGN\_REQUIRED
- CSV\_ESCALATE

- CSV\_FAIL
- CSV\_EMERGENCY\_PROVISIONAL
- CSV\_NOT\_MATERIAL

## Selectability rule

Option p is selectable for ordinary RLS ranking only if:

- RG supports the claim boundary,
- Rights Floor passes, formally NCRC(p) = PASS,
- TRC(p) = PASS,
- CSV\_status(p) is CSV\_PASS, CSV\_PASS\_WITH\_CONTROLS, or CSV\_NOT\_MATERIAL.

If CSV\_status(p) is CSV\_PASS\_WITH\_CONTROLS, the controls are part of the option. Removing them invalidates the pass and requires rerun.

If CSV\_status(p) is CSV\_NOT\_MATERIAL, the option is pass-equivalent for the declared claim boundary only after the PCC records why no material CSV burden is present. CSV\_NOT\_MATERIAL MUST NOT waive Structural Viability subchecks when execution feasibility, resource closure, dependency closure, reversibility, operational capacity, or internal coherence is material. If later evidence makes CSV material, CSV must be rerun.

If CSV\_status(p) is CSV\_REDESIGN\_REQUIRED or CSV\_FAIL, the option is not selectable as specified.

If CSV\_status(p) is CSV\_EMERGENCY\_PROVISIONAL, temporary action requires a time limit, harm cap, monitoring plan, review trigger, and transition/remediation plan. No unqualified alignment claim is permitted.

## CSV diagnostics

CSV may use internal concern levels from 0 to 4:

0 = no material concern. 1 = minor residual concern, carry to RLS. 2 = material concern, controls required. 3 = serious concern, redesign or escalation required. 4 = gate-failing concern.

Suggested diagnostic dimensions: containment integrity, structural viability, hollowing risk, dependency/lock-in risk, substitution pressure, mitigation adequacy, monitoring adequacy, reversibility/exit, accumulation risk, and legitimacy stress.

## UCI/HOI placement

UCI and HOI are not public cascade stages. They are first evaluated inside CSV when material to structural integrity. Residual UCI/HOI may be used only as tie-break, monitoring, or hollowing-risk documentation after RLS is tied, close, or non-decisive.

## CSV graduated verdict logic

CSV is a routing and selectability layer, not a purity filter. Its job is to distinguish ordinary bounded residual harm from uncontained or structurally non-viable harm.

- CSV\_PASS: contained and structurally viable; residuals may enter RLS.
- CSV\_PASS\_WITH\_CONTROLS: selectable only if specified controls become binding conditions.
- CSV\_REDESIGN\_REQUIRED: not selectable as specified, but a revised option may be evaluated.
- CSV\_ESCALATE: stakes, uncertainty, or authority gaps require deeper review before selection.
- CSV\_FAIL: uncontained, structurally degrading, unjustly externalized, non-viable, hidden, lock-in-producing, unmonitored, or beyond-tolerance harm.
- CSV\_EMERGENCY\_PROVISIONAL: temporary necessity-bounded action with time limit, harm cap, review trigger, and transition plan; not a full alignment certification.
- CSV\_NOT\_MATERIAL: no material CSV burden found for the declared claim boundary; pass-equivalent for selectability only after rationale is recorded. This status does not waive Structural Viability where execution feasibility is material.

## CSV diagnostic channels

CSV may use diagnostic ratings for containment integrity, structural viability, hollowing risk, dependency/lock-in risk, substitution pressure, mitigation adequacy, monitoring adequacy, reversibility, and accumulation risk. UCI/HOI-style diagnostics are evaluated inside CSV when material and are not independent public gates by default.

Formal selectability rule:  $A_{sel} = \{a \text{ in } A_{adm} : CSV\_status(a) \text{ in } \{CSV\_PASS, CSV\_PASS\_WITH\_CONTROLS, CSV\_NOT\_MATERIAL\}\}$ . CSV\_NOT\_MATERIAL is pass-equivalent only for the declared claim boundary after the PCC records the rationale. It does not waive Structural Viability where execution feasibility is material. If later evidence makes CSV material, CSV MUST be rerun. Emergency provisional options are handled through emergency protocol, not ordinary RLS selection.